

AREA AND PERIMETER OF PLANE FIGURES

Generally (7-9) questions have been asked from this chapter. Generally, questions are asked from the topics related to the triangles and circles.

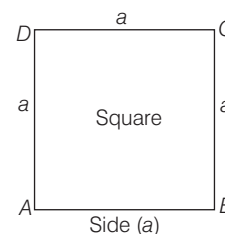


Area of the plane figure is the amount of surface enclosed by its boundary. It is measured in square units. The perimeter of plane figure is total length of the sides enclosing the figure. Unit of Perimeter is same as the unit of sides of a given figure.

SQUARE

Let each side of a square be a unit.

- (i) Perimeter of square = $4(\text{Side}) = 4a$ units
- (ii) Diagonal of square = $\sqrt{2} \times (\text{Side}) = a\sqrt{2}$ units
- (iii) Area of square = Side $\times$ Side = $(a)^2$ sq units = $\frac{1}{2}(\text{Diagonal})^2 = \frac{1}{2}d^2$ sq units



EXAMPLE 1. The perimeter of a square is $2(2x + 4y)$. Then, the area is

- a. $x^2 - 4xy + 4y^2$ b. $x^2 + 4y^2$ c. $x^2 - 4y^2$ d. $x^2 + 4xy + 4y^2$

Sol. d. Here, perimeter of a square = $2(2x + 4y)$

$$\therefore \text{Side of a square} = \frac{\text{Perimeter of a square}}{4} \therefore \text{Side of a square} = \frac{2(2x + 4y)}{4} = x + 2y$$

$$\therefore \text{Area of square} = (\text{Side})^2 \therefore \text{Area of square} = (x + 2y)^2 = x^2 + 4xy + 4y^2$$

EXAMPLE 2. The side of a square exceeds the side of the another square by 4 cm and the sum of the areas of the two squares is 400cm^2 . The dimensions of the squares are

- a. 8 cm and 12 cm b. 6 cm and 10 cm c. 12 cm and 16 cm d. None of these

Sol. c. Let the side of a square be x cm. Then, side of another square = $(x + 4)$ cm.

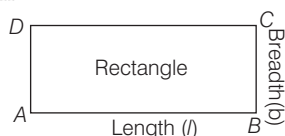
$$\text{Area of first square} = x^2 \text{ cm}^2 \text{ and area of second square} = (x + 4)^2 \text{ cm}^2$$

$$\text{According to the question} \Rightarrow x^2 + (x + 4)^2 = 400 \Rightarrow x^2 + x^2 + 16 + 8x = 400 \Rightarrow 2x^2 + 8x - 384 = 0$$

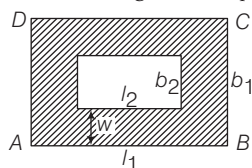
$$\begin{aligned}
 x^2 + 4x - 192 &= 0 \Rightarrow x^2 + 16x - 12x - 192 = 0 \\
 \Rightarrow x(x+16) - 12(x+16) &= 0 \Rightarrow (x-12)(x+16) = 0 \\
 \Rightarrow x-12 &= 0 \text{ or } x+16 = 0 \\
 \therefore x &= 12 \quad [\text{side cannot be negative}] \\
 \text{Hence, side of one square} &= 12 \text{ cm and side of another square} = 12 + 4 = 16 \text{ cm}
 \end{aligned}$$

RECTANGLE

Let l and b be the length and breadth of a rectangle respectively, then



- (i) Area of rectangle = Length $\times$ Breadth = $l \times b$
- (ii) Perimeter of rectangle = $2(\text{Length} + \text{Breadth})$
 $= 2(l + b)$ units
- (iii) Diagonal of rectangle = $\sqrt{(\text{Length})^2 + (\text{Breadth})^2}$
 $= \sqrt{l^2 + b^2}$ units
- (iv) Area of track (Shaded region) = $(l_1 b_1 - l_2 b_2)$ sq units



EXAMPLE 3. The area of a rectangular plot is 180 m^2 . If its length is 18 m. Then, its perimeter is

- a. 28 m b. 56 m c. 360 m d. None of these

Sol. b. Given, area of rectangular plot = 180 m^2

and length of the plot = 18 m

$$\therefore \text{Breadth} = \frac{\text{Area}}{\text{Length}} = \frac{180}{18} = 10 \text{ m}$$

Now, Perimeter of rectangular plot

$$= 2(\text{Length} + \text{Breadth}) = 2(18 + 10) = 56 \text{ m}$$

Hence, the perimeter of rectangular plot is 56 m.

EXAMPLE 4. The area of the floor of a rectangular hall of length 40 m is 960 m^2 . Carpets of size $6 \text{ m} \times 4 \text{ m}$ are available. Then, how many carpets are required to cover the hall?

- a. 20 b. 30 c. 40 d. 45

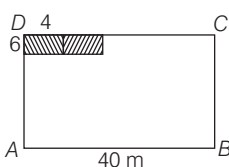
Sol. c. Given, area of the floor = 960 m^2

Area of one carpet = $6 \times 4 = 24 \text{ m}^2$

Number of carpets required

$$= \frac{\text{Area of the floor}}{\text{Area of one carpet}} = \frac{960}{24} = 40$$

Hence, 40 carpets are required to cover the hall.



EXAMPLE 5. A lawn is in the shape of a rectangle of length 60 m and width 40 m. There is a footpath of uniform width 1 m bordering the lawn. The area of the path is

- a. 194 m^2 b. 196 m^2 c. 198 m^2 d. 200 m^2

Sol. b. Given, length of the outer rectangle = 60 m and breadth of the outer rectangle = 40 m

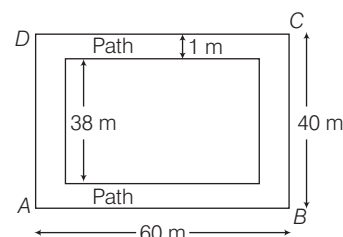
$$\therefore \text{Area of outer rectangle} = 60 \times 40 = 2400 \text{ m}^2$$

$$\therefore \text{Width of path} = 1 \text{ m}$$

$$\therefore \text{Length of the inner rectangle} = 60 - (1 + 1) = 58 \text{ m}$$

$$\text{Breadth of the inner rectangle} = 40 - (1 + 1) = 38 \text{ m}$$

$$\therefore \text{Area of the inner rectangle} = 58 \times 38 = 2204 \text{ m}^2$$



$$\text{Area of the path} = [\text{Area of the outer rectangle}]$$

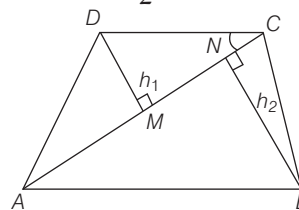
$$- [\text{Area of the inner rectangle}] = (2400 - 2204) = 196 \text{ m}^2$$

Hence, the area of the path is 196 m^2 .

QUADRILATERAL

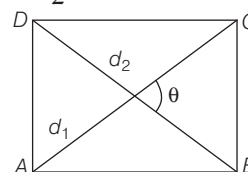
- (i) Let $ABCD$ is a quadrilateral in which $DM = h_1$ and $BN = h_2$ are perpendiculars on diagonal AC from other two vertices B and D , then

$$\begin{aligned}
 \text{Area of the quadrilateral} &= \frac{1}{2} \times \text{Diagonal} \times (h_1 + h_2) \\
 &= \frac{1}{2} \times AC \times (DM + BN) \text{ sq units}
 \end{aligned}$$



- (ii) Let d_1 and d_2 are two diagonals and θ is angle between them, then area of quadrilateral

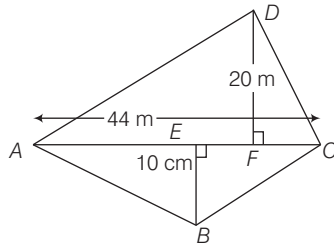
$$= \frac{1}{2} d_1 d_2 \sin \theta$$



EXAMPLE 6. In a quadrilateral $ABCD$, diagonal $AC = 44$ cm and the length of the perpendicular drawn from B and D to AC are 10 cm and 20 cm, respectively. The area of the quadrilateral is

- a. 330 cm^2 b. 440 cm^2 c. 550 cm^2 d. 660 cm^2

Sol. d.



Area of quadrilateral

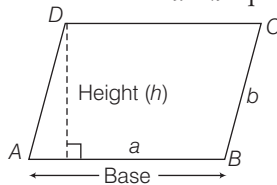
$$= \frac{1}{2} AC(h_1 + h_2) = \frac{1}{2} (44) (20 + 10) = \frac{1}{2} \times 44 \times 30 = 660 \text{ cm}^2$$

Hence, the area of the quadrilateral is 660 cm^2 .

PARALLELOGRAM

Let adjacent sides of a parallelogram are b and a respectively and h is the corresponding altitude (height) of side a , then

- (i) Area of the parallelogram = (Base $\times$ Height)
 $= a \times h$ sq units



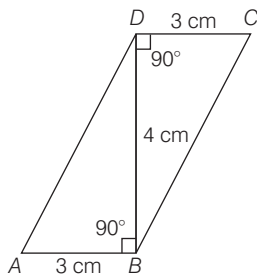
- (ii) Perimeter of a parallelogram = $2(\text{Sum of adjacent side})$
 $= 2(a + b)$ units

Note Each diagonal of a parallelogram divides it into two triangles of equal area.

EXAMPLE 7. $ABCD$ is a parallelogram as shown in figure, then its area is

- a. 12 cm^2 b. 14 cm^2 c. 15 cm^2 d. 18 cm^2

Sol. a. Given, base = 3 cm and height = 4 cm



Area of parallelogram,

$$ABCD = \text{Base} \times \text{height} = 3 \times 4 = 12 \text{ cm}^2$$

EXAMPLE 8. In the parallelogram $ABCD$, $AB = 10$ cm. The altitude corresponding to sides AB and AD are respectively 7 cm and 8 cm. Then, AD is

- a. 8.75 cm b. 8.95 cm c. 9 cm d. 9.25 cm

Sol. a. Area of parallelogram $ABCD$

= Base $\times$ Corresponding altitude

$\therefore$ Area of parallelogram

$$ABCD = AB \times DM$$

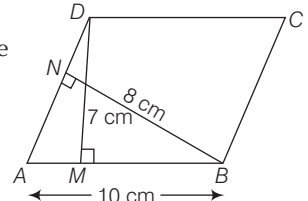
$$= 10 \times 7 = 70 \text{ cm}^2 \quad \dots(i)$$

Also, area of parallelogram

$$ABCD = AD \times BN = 8AD \quad [\because BN = 8 \text{ cm}] \quad \dots(ii)$$

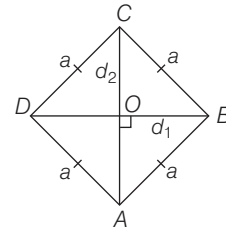
From Eqs. (i) and (ii), $8AD = 70$

$$\therefore AD = \frac{70}{8} = 8.75 \text{ cm}$$



RHOMBUS

Let the length of each sides of a rhombus is a and length of both diagonals are d_1 and d_2 , then



- (i) Area of rhombus = $\frac{1}{2} \times d_1 \times d_2$ sq units

$$(ii) \text{ Side of rhombus} = \frac{1}{2} \sqrt{d_1^2 + d_2^2} \text{ units} \Rightarrow 4a^2 = d_1^2 + d_2^2$$

- (iii) Perimeter of rhombus = $4 \times (\text{Side})$ units

Note Diagonals of a rhombus bisect each other at right angles.

EXAMPLE 9. If the length of the diagonal of a rhombus is $(a + b)$ and its area is $\frac{a^2 - b^2}{2}$ sq units, then the other diagonal is

- a. $a + b$ b. $a - b$ c. $\frac{a - b}{2}$ d. $\frac{a + b}{2}$

Sol. b. Let second diagonal be d units.

$$\text{Given, area of rhombus} = \frac{a^2 - b^2}{2}$$

$$\therefore \text{Area of rhombus} = \frac{1}{2} \times d_1 \times d_2$$

$$\therefore \frac{1}{2} \times d_1 \times d_2 = \frac{a^2 - b^2}{2} \Rightarrow \frac{1}{2} (a + b) \cdot d = \frac{a^2 - b^2}{2}$$

$$[\because d_1 = a + b]$$

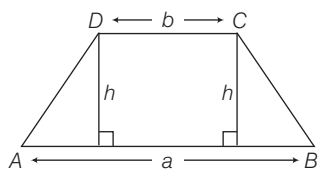
$$\Rightarrow (a + b) d = a^2 - b^2 \Rightarrow d = \frac{(a^2 - b^2)}{a + b} = \frac{(a - b)(a + b)}{(a + b)}$$

$$\therefore d = (a - b) \quad [\because a^2 - b^2 = (a - b)(a + b)]$$

TRAPEZIUM

Let the length of parallel sides of a trapezium are a and b and distance between them is h , then

$$\text{Area of trapezium} = \frac{1}{2} (\text{Sum of parallel sides}) \times (\text{Distance between them})$$



$$= \frac{1}{2} (AB + CD) \times h = \frac{1}{2} (a + b) \times h \text{ sq units}$$

EXAMPLE 10. The difference between two parallel sides of a trapezium is 4 cm and the perpendicular distance between them is 19 cm. Find the lengths of the parallel sides, if the area of the trapezium is 475 cm^2 .

- a. 22 cm and 18 cm b. 25 cm and 21 cm
c. 29 cm and 25 cm d. 27 cm and 23 cm

Sol. d. Let the length of the parallel sides of the trapezium be a cm and b cm, respectively.

Then, according to the question, $(a - b) = 4$... (i)

$$\therefore \text{Area of trapezium} = \frac{1}{2} \times (a + b) \times h$$

$$\frac{1}{2} \times (a + b) \times 19 = 475$$

$$\Rightarrow a + b = \frac{950}{19} = 50$$

$$\therefore a + b = 50 \quad \dots \text{(ii)}$$

On solving Eqs. (i) and (ii), we get $a = 27$ and $b = 23$

Thus, length of parallel sides are 27 cm and 23 cm.

SCALED TRIANGLE

Let the sides of a triangle are a, b, c and h be the corresponding altitude to side a , then

(i) Perimeter of scalene triangle, $2s = (a + b + c)$ units

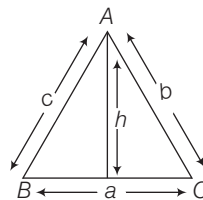
(ii) Semi-perimeter of scalene triangle, $s = \frac{a + b + c}{2}$ units

(iii) Area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$ sq units

$$\text{where, } s = \frac{a + b + c}{2} \quad [\text{Heron's formula}]$$

$$\text{or area of triangle} = \frac{1}{2} \times a \times h \text{ sq units}$$

(iv) The radius of incircle = $\frac{2(\text{Area})}{\text{Perimeter}}$



(v) The radius of circumcircle = $\frac{abc}{4(\text{Area})}$

EXAMPLE 11. The area of a triangle whose sides are 9 cm, 12 cm and 15 cm is

- a. 45 cm^2 b. 54 cm^2 c. 56 cm^2 d. 64 cm^2

Sol. b. Here $a = 9$ cm, $b = 12$ cm and $c = 15$ cm

$$s = \frac{a + b + c}{2} = \frac{9 + 12 + 15}{2} = 18 \text{ cm}$$

$$\text{Area of a triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

[Heron's formula]

$$= \sqrt{18(18-9)(18-12)(18-15)}$$

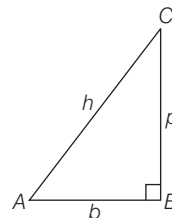
$$= \sqrt{18 \times 9 \times 6 \times 3} = 54 \text{ cm}^2$$

Hence, the area of a triangle is 54 cm^2 .

RIGHT ANGLED TRIANGLE

A figure bounded by three straight lines is called a triangle.

Let perpendicular, base and hypotenuse of a right angled triangle ($\triangle ABC$) be p, b and h , respectively, then



(i) Perimeter of right angled triangle = $AB + BC + CA$
 $= (b + p + h)$ units

(ii) Area of right angled triangle = $\frac{1}{2} \times \text{Base} \times \text{Altitude}$
 $= \frac{1}{2} \times b \times p$ sq units

EXAMPLE 12. The base of triangular field is three times its altitude. If the cost of cultivating the field at 50 per hectare be ₹ 675, then its base and height are

- a. 900 m and 300 m b. 600 m and 300 m
c. 500 m and 200 m d. None of these

Sol. a. Area of the triangular field = $\frac{\text{Total cost}}{\text{Rate}}$

$$= \frac{675}{50} = 13.5 \text{ hec}$$

$$= (13.5 \times 10000) \text{ m}^2 = 135000 \text{ m}^2 \quad \dots \text{(i)}$$

Let altitude of triangular field be x m

and base of field = $3x$ m

Again, area of the field = $\frac{1}{2} \times \text{Base} \times \text{Altitude}$

$$= \frac{1}{2} \times 3x \times x = \frac{3x^2}{2} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{3x^2}{2} = 135000$$

$$\therefore x^2 = \frac{135000 \times 2}{3} = 90000 \Rightarrow x = 300 \text{ m}$$

So, base = $3x = 3 \times 300 = 900$ m

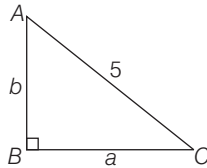
and altitude = $x = 300$ m

EXAMPLE 13. The perimeter of a right triangle is 12 cm. The hypotenuse is 5 cm. The other two sides and area of the triangle are

- a. 3, 4 and 6 cm^2 b. 4, 3 and 12 cm^2
c. 6, 2 and 6 cm^2 d. None of these

Sol. a. Let other two sides of a right triangle be a cm and b cm such that $a > b$.

Given, perimeter of a right angled triangle = 12 cm



$$\therefore a + b + 5 = 12 \Rightarrow a + b = 7 \text{ cm} \quad \dots(i)$$

Also, by pythagoras theorem,

$$a^2 + b^2 = 25$$

Also, $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$

$$(a - b)^2 = 2(a^2 + b^2) - (a + b)^2$$

$$\Rightarrow (a - b)^2 = 2(25) - (7)^2 = 50 - 49 = 1$$

$$\therefore a - b = 1 \quad [\because a > b] \dots(ii)$$

On solving Eqs. (i) and (ii), we get

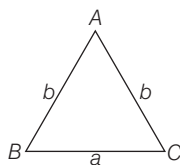
$$a = 4 \text{ cm and } b = 3 \text{ cm}$$

$$\therefore \text{Area of triangle} = \frac{1}{2} \times \text{Base} \times \text{Altitudes}$$

$$\Rightarrow \frac{1}{2} \times 3 \times 4 = 6 \text{ cm}^2$$

ISOSCELES TRIANGLE

Let sides of an isosceles triangle are a , b and b , then



(i) Perimeter of isosceles triangle
= $a + b + b = a + 2b$ units

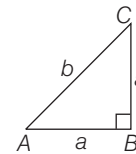
(ii) Area of isosceles triangle

$$= (s - b)(\sqrt{s(s - a)}) = \frac{1}{4} a \sqrt{4b^2 - a^2} \text{ sq units}$$

where, a = Base and b = Equal sides and
 s = semi-perimeter

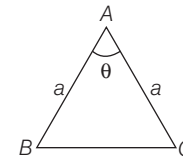
(iii) Area of a right isosceles triangle, in which equal sides from a right angle is given by

$$\text{Area} = \frac{1}{2} a^2 \text{ sq units}$$



(iv) If θ is the angle between two equal sides of isosceles triangle, then

$$\text{Area of triangle} = \frac{1}{2} a^2 \sin \theta$$



EXAMPLE 14. The perimeter of an isosceles triangle is 32 cm while equal sides together measure 20 cm. Then, area of an isosceles triangle is

- a. 48 cm^2 b. 84 cm^2 c. 44 cm^2 d. 41 cm^2

Sol. a. Let third side of an isosceles triangle be a cm.

Given, $b = 10$ cm

$$\therefore \text{Perimeter of isosceles triangle} = a + b + b = a + 2b$$

$$\Rightarrow 32 = a + 2 \times 10 = a + 20 \Rightarrow 32 - 20 = a$$

$$\therefore a = 12 \text{ cm}$$

Now, area of an isosceles triangle

$$= \frac{1}{4} a \sqrt{4b^2 - a^2} = \frac{1}{4} \times 12 \sqrt{4 \times 10^2 - 12^2}$$

$$= 3\sqrt{400 - 144} = 3 \times 16 = 48 \text{ cm}^2$$

EQUILATERAL TRIANGLE

Let a be the side of an equilateral triangle, then

(i) Height (altitude) of an equilateral triangle = $\frac{\sqrt{3}}{2} a$

(ii) Area of an equilateral triangle = $\frac{\sqrt{3}}{4} a^2$

(iii) Perimeter of an equilateral triangle = $3 \times \text{Side} = 3a$

(iv) The radius of incircle = $\frac{a}{2\sqrt{3}}$